Simulation-based Time Series Analysis:

Testing for Structure, Likelihood-free Estimation, and Prediction with Transformers using Synthetic Data

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Overview of presentation

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Part I

Testing

Economics application

Hardness result

Physical dependence

Type I error control

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Part II

Estimation

Random features

Embedding results

Consistency

Asymptotic normality

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Part III

Prediction

Foundation models

Sim-and-regression

Conf. distributions

Calibration

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- **Temporal dependence:** X_t may depend on the past $X_{t-1}, X_{t-2}, \ldots$
- **Nonstationarity:** the joint distribution of $X_{t_1+\tau}, \ldots, X_{t_m+\tau}$ at $m \in \mathbb{N}$ distinct time points $t_1, \ldots, t_m$ may change with the time-offset τ

Conditional independence testing

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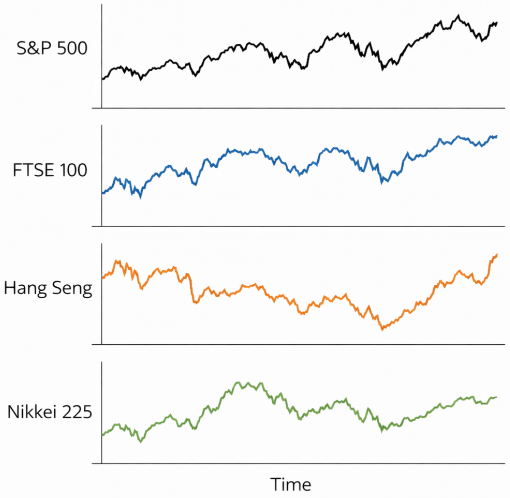
- **Example:** links in the global economic system

Real data application: links in the global economic system

- Four stock market indices:
  1. S&P 500 (US)
  2. FTSE 100 (UK)
  3. Hang Seng (HK)
  4. Nikkei 225 (JP)
- Daily returns based on closing prices
- Trading days Jan. 2022–Apr. 2025
- Sample size: $n = 860$
- Results based on BH-adjusted p-values



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- Next, we present the testing procedure

Algorithm for conditional independence test (1/2)

1. **Inputs:** Data $X_{1:n}$, $Y_{1:n}$, $Z_{1:n}$, significance level α, number of simulations s, $p \in [2, \infty]$ for norm in test statistic, and method for selecting the window size L

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5. Calculate test statistic on time series of residual products

$$T(\hat{R}_{1:n}) = \max_{T \in [n]} \left\| \frac{1}{\sqrt{n}} \sum_{t \leq T} \hat{R}_t \right\|_p$$

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11. Reject null hypothesis at level α if $T(\hat{R}_{1:n}) > \hat{q}_{1-\alpha}^{\mathrm{boot}}$, else retain

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- Assume this for all time series $X_{1:n}$, $Y_{1:n}$, $Z_{1:n}$

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- **We do not know** the generative model $G_t^{(n)}$, noise inputs $\boldsymbol{\varepsilon}_t$, or parameter θ

Assumption 2: Decaying influence of noise inputs (W. B. Wu 2005)

- Let $(\varepsilon_i^*)_{i\in\mathbb{Z}}$ be an iid copy of $(\varepsilon_i)_{i\in\mathbb{Z}}$. For $\ell \in \mathbb{Z}$, $j \in \mathbb{N}_0$, denote

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Assumption

There exist constants $C > 0$, $\rho > 1$ such that, for some $q > 2$ and all $j \in \mathbb{N}$, we have

$$\sup_{i,t,\ell,\theta} \left\| G_t^{(i)}(\boldsymbol{\varepsilon}_\ell, \theta) - G_t^{(i)}(\tilde{\boldsymbol{\varepsilon}}_{\ell,j}, \theta) \right\|_{L^q(\theta)} \leq C(j \vee 1)^{-\rho}$$

where $\|\cdot\|_{L^q(\theta)} = \mathbb{E}_\theta\left(\|\cdot\|_2^q\right)^{1/q}$, $\mathbb{E}_\theta(\cdot)$ *is expectation w.r.t.* θ*-dependent distribution*

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Theorem (Informal)

Suppose Assumptions 1-3 hold for all distributions $P \in \mathcal{P}_n^* \subset \mathcal{P}_n^{\mathrm{CI}}$ for every $n \in \mathbb{N}$. Then we have uniformly asymptotic Type I error control

$$\limsup_{n \to \infty} \sup_{P \in \mathcal{P}_n^*} \mathbb{P}_P \left(T(\hat{R}_{1:n}) > \hat{q}_{1-\alpha}^{\mathrm{boot}} \right) \leq \alpha$$

where $T(\hat{R}_{1:n})$ is the test statistic and $\hat{q}_{1-\alpha}^{\mathrm{boot}}$ is the estimated quantile.

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- The likelihood is intractable, so we use a simulation-based approach
- **Assumption:** We can simulate from an *approximation* $G_t^{(i)}$ to the true mapping G_t at any value of $\theta \in \Theta$, where the approximation improves as $i \to \infty$
  – Note: Allows us to consider growing burn-in periods, discretized ODEs, etc

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- Details: the $(2p+1) \times p$ Jacobian matrix of Φ has full rank p on each compact subset of a smooth manifold contained in $\mathbb{R}^p$

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- **Key idea:** Estimate θ_0 by minimizing distance of $\hat{\Phi}(\theta_0)$ and $\hat{\Phi}(\theta)$ over $\theta \in \Theta$, $\hat{\Phi}(\theta)$ is a time-average of $2p+1$ random features of a time series generated at θ

Example: Mean of Gaussian

- **Goal:** Estimate $\theta_0 = \mu_0 = 0.5$ given $X_1^{\text{obs}}, \ldots, X_n^{\text{obs}} \overset{\text{iid}}{\sim} \mathcal{N}(\theta_0, 1)$ with $\Theta = [-3, 3]$

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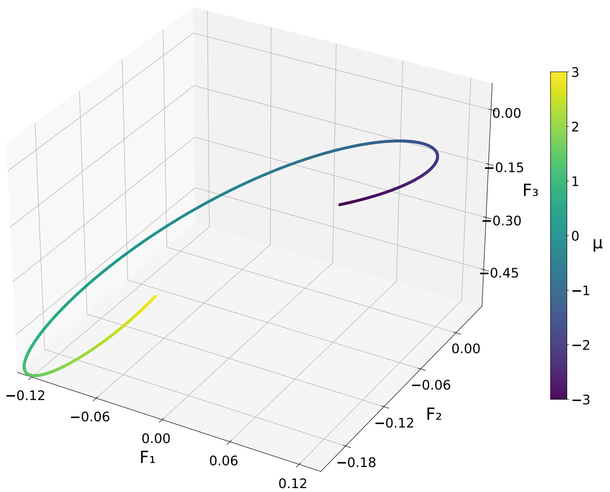
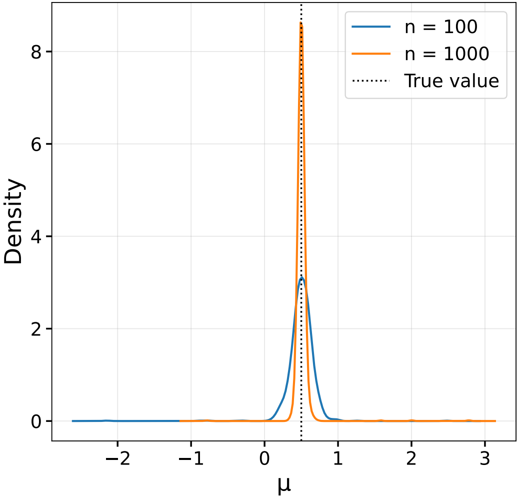
- Let's visualize an embedding and check the estimation results

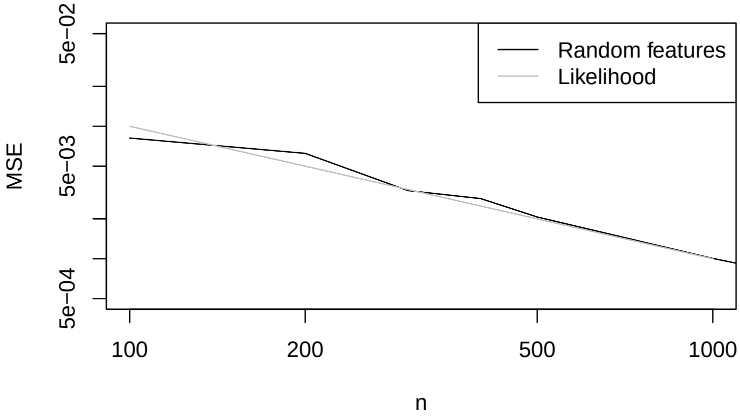
Each point is a time-average of 3 features, colored by value of parameter μ



Density plot from $1,000$ independent estimates (different random features)



Theoretical MSE of $\hat{\theta}^{\mathrm{MLE}}$ $(1/n)$ vs MSE of our $\hat{\theta}$ (100 replicates per n)



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  1. Adaptively fine-tuning pre-trained transformers with synthetic data
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Paradigm shift in prediction: Time series foundation models (TSFMs)

- Inspired by success of transformer-based foundation models in language and vision
- Many TSFMs are *pre-trained* on synthetic data generated by *parametric* models
- Renewed interest in finding parametric models that generate realistic time series
- X. Liu et al. (2025) argue that "adaptively fine-tuning" TSFMs on "carefully designed synthetic data" remains a "missed opportunity"
- Statistical foundations are unclear, especially for nonstationary time series
- We develop a method that can be used for:
  1. Adaptively fine-tuning pre-trained transformers with synthetic data
  2. A robust form of simulation-and-regression with untrained transformers
- To begin, we explain classical simulation-and-regression methods

Example to keep in mind: State-space model

- Decomposes observed time series into latent states and noise

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where M_t is a measurement function and S_t is a state transition function

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- We can also consider control inputs Z_t, $t = 1, \dots, n$
- **Smoothing:** estimate past latent states with current information
- **Forecasting:** predict future observations or states with current information

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- But $\hat{\theta} \neq \theta_0$, so the minimizer $f^*_{P_{\hat{\theta}}}$ in the function class of $R_{P_{\hat{\theta}}}(f)$ will be "wrong"

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  1. Classical simulation-and-regression when $\widehat{\mathrm{CD}}_n$ is Dirac measure at $\hat{\theta}$
  2. TSFM pre-training with synthetic data when $\widehat{\mathrm{CD}}_n$ is uniform over Θ

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- **Primary objectives until defense:** Develop method, evaluate performance with extensive forecasting and smoothing experiments, write paper, and submit paper
- *After submitting the main paper*, if time allows, some possible extensions include:
  - Pre-training vs training from scratch
  - Parameter confidence distribution vs parameter point estimate
  - Application to epidemiology

Thank you!

Questions and comments are welcome

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Carnegie Mellon University

Main ideas

- **Research question:** Does incorporating parameter uncertainty improve the calibration of pre-trained/untrained transformers that output quantile predictions?
- We would like the quantile predictions to have the correct coverage
- That is, the probability that the time series, at a particular time, is less than or equal to the predicted α-quantile, at that same time, is *actually* equal to α
- For smoothing, we assess this with synthetic experiments
- For forecasting, we assess this with real data and synthetic experiments

Multi-quantile loss function

- Denote the times by $\mathcal{T}$, and number of times by $T = |\mathcal{T}|$, dimensions by d, quantile levels by K, and simulations by s (in practice, use batches)
- Multi-quantile regression heads by minimizing sum of multiple quantile losses

$$\frac{1}{s} \sum_{r \in [s]} \left[\sum_{t \in \mathcal{T}} \left(\sum_{j \in [d]} \left[\sum_{k \in [K]} \rho_{\tau_k} \left(X_{t,j}^{(r)} - [f(Y^{(r)})]_{t,j,\tau_k} \right) + Q_\lambda(r,t,j) \right] \right) \right]$$

  where $\rho_{\tau_k}(u) = (\tau_k - \mathbb{I}[u < 0])\, u$, each $\tau_k \in (0,1)$, and $\mathbb{I}$ is the indicator function
- Want predicted quantiles at lower levels to be less than those at higher levels, i.e. denoting $\hat{q}_{t,j}^{\mathrm{obs}}(\tau_k) \equiv [\hat{f}(Y^{\mathrm{obs}})]_{t,j,\tau_k}$ we want $\hat{q}_{t,j}^{\mathrm{obs}}(\tau_1) \leq \hat{q}_{t,j}^{\mathrm{obs}}(\tau_2) \leq \ldots \leq \hat{q}_{t,j}^{\mathrm{obs}}(\tau_K)$, so we penalize **quantile crossing** by adding the penalty

$$Q_\lambda(r,t,j) = \lambda \sum_{k=1}^{K-1} \max\left(0, [f(Y^{(r)})]_{t,j,\tau_k} - [f(Y^{(r)})]_{t,j,\tau_{k+1}}\right)$$

  for some tuning parameter $\lambda > 0$, where the levels are ordered $\tau_1 \leq \tau_2 \leq \ldots \leq \tau_K$

Algorithm for calibration experiments

1. Estimate confidence distribution $\widehat{\mathrm{CD}}_n$ using the observed time series Y^{obs}
2. For $r \in [s]$, draw $\theta^{(r)} \sim \widehat{\mathrm{CD}}_n$ and simulate $(X^{(r)}, Y^{(r)})_{r\in[s]}$ at $\theta^{(r)}$
3. Select hyperparameters (e.g., K-fold or $80/20$ train/val over simulations)
4. For multi-quantile loss L, obtain predictor $\hat{f}$ by minimizing

   $$\frac{1}{s}\sum_{r\in[s]}\left[\sum_{t\in\mathcal{T}} L\left(X_t^{(r)} - [f(Y^{(r)})]_t\right)\right]$$

   over some seq2seq function class (e.g., transformers)
5. Apply seq2seq function $\hat{f}(Y^{\mathrm{obs}})$ to predict latent states at all times based on the multiple predicted quantiles

Setup for calibration metrics

- To assess calibration in the context of quantile forecasting nonstationary time series, we adapt the metrics from Adler et al. (2025), which use time-averages
- **Main difference:** We average each metric over s simulations from a model
- When forecasting observations of real time series, we can use time-averages
- When forecasting observations, just let X denote the forecasting target
- We present the metrics for the univariate setting, i.e. $d = 1$, to simplify the notation, and for $d \in \mathbb{N}$, we can aggregate metrics by averaging over dimensions
- Let $\mathcal{T}$ to be the times with $T = |\mathcal{T}|$, Y be the observed time series, $\hat{q}_t^{(r)}(\tau_k) \equiv [\hat{f}(Y^{(r)})]_{t,\tau_k}$ be the r-th predicted quantile at time t for level $\tau_k \in \{0.1, 0.2, \ldots, 0.9\}$, and $\Delta = \{(0.1, 0.9), (0.2, 0.8), (0.3, 0.7), (0.4, 0.6)\}$

Calibration metric 1

- For $t \in \mathcal{T}$, define the *Probabilistic Calibration Error (PCE)* as:

$$\mathrm{PCE}(t) = \frac{1}{K} \sum_{k \in [K]} \left| \tau_k - \frac{1}{s} \sum_{r=1}^{s} \mathbb{I}\left[X_t^{(r)} \leq \hat{q}_t^{(r)}(\tau_k) \right] \right|$$

- Lower values indicate better-calibrated models:
  1. PCE is lower-bounded by 0 (i.e. predicted quantile above)
  2. PCE is upper-bounded by 0.5 (i.e. predicted quantile below)

Calibration metric 2

- Well-calibrated models do not necessarily yield useful forecasts
- A model can predict very large quantiles and be calibrated
- Need metric that captures sharpness (i.e. concentration of predictive distribution)
- Scale width of prediction interval by width of corresponding interval using unconditional quantiles $\bar{q}_t^{(r)}(\cdot)$ estimated via order statistics of s simulations
- For $t \in \mathcal{T}$, define the *Scaled Interval Width (SIW)* as:

$$\mathrm{SIW}(t) = \frac{1}{|\Delta|} \sum_{(\tau_\ell, \tau_h) \in \Delta} \left[\frac{1}{s} \sum_{r=1}^{s} \frac{\hat{q}_t^{(r)}(\tau_h) - \hat{q}_t^{(r)}(\tau_\ell)}{\bar{q}_t^{(r)}(\tau_h) - \bar{q}_t^{(r)}(\tau_\ell)} \right]$$

- Lower (resp. higher) values correspond to tighter (resp. wider) prediction intervals

Calibration metric 3

- We introduce a metric to help check whether a model is under or over confident
- For $t \in \mathcal{T}$, define the *Centered Calibration Error (CCE)* as:

$$\mathrm{CCE}(t) = \frac{1}{|\Delta|} \sum_{(\tau_\ell, \tau_h) \in \Delta} \left((\tau_h - \tau_\ell) - \frac{1}{s} \sum_{r=1}^{s} \mathbb{I}\left[\hat{q}_t^{(r)}(\tau_\ell) \leq X_t^{(r)} \leq \hat{q}_t^{(r)}(\tau_h) \right] \right)$$

- Positive CCE values mean that more observations are outside the prediction interval than expected
- When combined with a low SIW value, this suggests a model is overconfident, i.e. that the prediction intervals are too narrow

Calibration

- For improving calibration with $\widehat{\mathrm{CD}}_n$ *beyond* doing simulation-and-regression and fine-tuning based on the parameter estimate $\hat{\theta}$, we can reduce this to the other questions by instantiating the loss function L as the multi-quantile loss
- For untrained transformers, refer to *simulation-based training*
- For pre-trained transformers, refer to *pre-training*

Simulation-based training

- **Question:** In the context of simulation-and-regression, does accounting for parameter uncertainty achieve better performance than using parameter estimate?
- When does this inequality hold:

$$\mathbb{E}_{P_{\theta_0}}\left[\sum_{t\in\mathcal{T}} L\left(X_t - [f^*_{P_{\widehat{\text{mix}}}}(Y)]_t\right)\right] < \mathbb{E}_{P_{\theta_0}}\left[\sum_{t\in\mathcal{T}} L\left(X_t - [f^*_{P_{\hat{\theta}}}(Y)]_t\right)\right]$$

where both expectations are with respect to the distribution P_{θ_0} of the time series corresponding to the unknown true parameter θ_0, $f^*_{P_{\widehat{\text{mix}}}}$ is the minimizer in the function class of $R_{P_{\widehat{\text{mix}}}}(f)$, and $f^*_{P_{\hat{\theta}}}$ is the minimizer in the function class of $R_{P_{\hat{\theta}}}(f)$

Pre-training

- **Question:** When does a pre-trained transformer fine-tuned with our method outperform a transformer trained from scratch with our method?
- Intuitively, this occurs when the following conditions hold
- Large parameter uncertainty, e.g., when the sample size of the observed time series is small relative to the noise
- The distributions used in the pre-training phase are close to the true distribution, e.g., when we have domain knowledge or we have previously observed time series with similar distributions

Connections of proposed method to Bayesian prediction

- When using the Gaussian depth-CD $N(\hat{\theta}, \hat{\Sigma}/n)$, our proposed method can be viewed as a frequentist, likelihood-free analogue of the Laplace approximation to the posterior distribution in Bayesian statistics
- Specifically, when using the Laplace approximation to the posterior distribution to obtain an approximate posterior predictive distribution
- Recall that the Laplace approximation uses a Gaussian distribution centered at the maximum a posteriori (MAP) estimate with covariance given by the inverse observed Fisher information to approximate the posterior distribution
- The Bernstein–von Mises theorem is used to justify the Laplace approximation
- In contrast, to justify our approach, we use asymptotic theory for likelihood-free parameter estimators for parametric time series models, nonstationary time series theory (Mies 2023), and the theory of confidence distributions

Definition of confidence distribution

- In general, our frequentist procedure can be used with any *confidence distribution* over $\Theta \subset \mathbb{R}^p$ which expresses our uncertainty about the $\theta_0 \in \mathbb{R}^p$
- Ideally, this distribution is easy to simulate from
- We give the definition of confidence distributions for the $p = 1$ case, and the generalization to the $p \in \mathbb{N}$ case
- In the definition, $\Theta \subset \mathbb{R}^p$ is the parameter space and $\mathcal{X}_n$ is the sample space corresponding to the data $X_{1:n}$

Definition of confidence distribution

Definition (Confidence Distribution, $p = 1$, (Min-ge Xie and Singh 2013))

A function $\widehat{\mathrm{CD}}_n(\cdot) = \widehat{\mathrm{CD}}_n(X_{1:n}, \cdot)$ on $\mathcal{X}_n \times \Theta \to [0,1]$ is called a confidence distribution (CD) for a parameter θ, if:

1. For each given $X_{1:n}$, $\widehat{\mathrm{CD}}_n(\cdot) = \widehat{\mathrm{CD}}_n(X_{1:n}, \cdot)$ is a CDF on Θ
2. At the true parameter value $\theta = \theta_0$, $\widehat{\mathrm{CD}}_n(\theta_0) \equiv \widehat{\mathrm{CD}}_n(X_{1:n}, \theta_0)$, as a function of the sample $X_{1:n}$, follows the uniform distribution $U[0,1]$

- Also, the function $\widehat{\mathrm{CD}}_n(\cdot)$ is an asymptotic confidence distribution (aCD), if the $U[0,1]$ requirement is true only asymptotically

Definition of confidence distribution

- The definition of confidence distributions for the general $p \in \mathbb{N}$ case involves depth functions
- As discussed in R. Y. Liu, Parelius, and Singh (1999), a data depth is a way of measuring how deep or central a point $x \in \mathbb{R}^d$ is with respect to a probability distribution P on $\mathbb{R}^d$, $d \in \mathbb{N}$, or with respect to a given data cloud $X_{1:n}$
- In general, a point with a larger depth value means that it is closer to the center of the distribution or more central in a data cloud, while points with smaller depth values are farther away from the center, so data depth is a center-outward ordering of multivariate observations

Definition of confidence distribution

- For example, the Mahalanobis depth (D_M) (Mahalanobis 1936) at $x \in \mathbb{R}^d$ with respect to P is defined as

$$D_M(P, x) = \frac{1}{1 + (x - \mu_P)^\top \Sigma_P^{-1} (x - \mu_P)},$$

  where μ_P and Σ_P are the mean vector and the dispersion matrix of P, respectively
- We obtain the sample version of D_M by replacing μ_P and Σ_P by the corresponding sample estimates $\hat{\mu}_P$ and $\hat{\Sigma}_P$
- See R. Y. Liu, Parelius, and Singh (1999) for more examples of data depth

Definition of confidence distribution

Definition (Confidence Distribution, $p \in \mathbb{N}$, (D. Liu, R. Y. Liu, and Min-ge Xie 2022))

A function $\widehat{\mathrm{CD}}_n(\cdot) \equiv \widehat{\mathrm{CD}}_n(X_{1:n}, \cdot)$ on $\Theta \subset \mathbb{R}^p$ is called a depth confidence distribution (depth-CD) associated with depth function D for a vector-valued parameter θ, if:

1. It is a distribution function on the parameter space Θ for any fixed sample set $X_{1:n}$
2. The $(1-\alpha)$ "central region" of the distribution $\widehat{\mathrm{CD}}_n(\cdot)$,

$$\mathcal{R}_{1-\alpha}(\widehat{\mathrm{CD}}_n) = \{\theta \in \Theta : C_{\widehat{\mathrm{CD}}_n}(\theta) \geq \alpha\},$$

   is a confidence region for θ with a coverage probability of $(1-\alpha)$. Here, the centrality function associated with depth D and confidence distribution $\widehat{\mathrm{CD}}_n(\cdot)$, that is, $C_{\widehat{\mathrm{CD}}_n}(\theta)$, is also referred to as a depth confidence curve (depth-CV)

- If the statements in (2) hold only asymptotically, then we refer to $\widehat{\mathrm{CD}}_n$ and $C_{\widehat{\mathrm{CD}}_n}$ as asymptotic depth-CD and asymptotic depth-CV, respectively

Depth-CV for Gaussian depth-CD

- In this work, we focus on asymptotic Gaussian depth confidence distributions of the form $\widehat{\mathrm{CD}}_n = N(\hat{\theta}, \hat{\Sigma}/n)$ for a multi-dimensional parameter $\theta \in \mathbb{R}^p$, $p \in \mathbb{N}$
- Let $q_{1-\alpha}$ be the $1-\alpha$ quantile of χ^2_p, i.e., $\mathbb{P}(\chi^2_p \leq q_{1-\alpha}) = 1-\alpha$
- As usual, the asymptotic $1-\alpha$ confidence region for θ_0 is

$$\mathcal{R}_{1-\alpha}(\widehat{\mathrm{CD}}_n) = \{\theta \in \Theta : (\theta - \hat{\theta})^\top (\hat{\Sigma}/n)^{-1} (\theta - \hat{\theta}) \leq q_{1-\alpha}\}$$

so applying the χ^2_p CDF $F_{\chi^2_p}$, subtracting by 1, dividing by -1, yields

$$\mathcal{R}_{1-\alpha}(\widehat{\mathrm{CD}}_n) = \{\theta \in \Theta : C_{\widehat{CD}_n}(\theta) = 1 - F_{\chi^2_p}[(\theta - \hat{\theta})^\top (\hat{\Sigma}/n)^{-1} (\theta - \hat{\theta})] \geq \alpha\}$$

Result for confidence distributions

- We justify this Gaussianity through the asymptotic normality of the parameter estimator, nonstationary time series theory (Mies 2023), and the following result from D. Liu, R. Y. Liu, and Min-ge Xie (2022) (Theorem 2 therein)

Theorem (Depth-CDs from Pivot Statistics (D. Liu, R. Y. Liu, and Min-ge Xie 2022))

Assume that $A_n(X_{1:n})$ is a nonsingular matrix such that $A_n(X_{1:n})(\hat{\theta} - \theta)$ follows a distribution Q_n that is free of all unknown parameters. Also assume that η_n is a random vector, independent of the sample $X_{1:n}$, following the distribution Q_n. Then, conditional on $X_{1:n}$, the distribution function of $(\hat{\theta} - A_n(X_{1:n})^{-1}\eta_n)$ is a depth-CD for θ, under the following conditions:

1. *The depth D is affine-invariant.*
2. *The depth contours $\{\eta_n : D_{Q_n}(\eta_n) = \epsilon\}$ all have probability zero with respect to Q_n for any $\epsilon > 0$.*

Bootstrap distribution is a confidence distribution

- Alternatively, use parametric bootstrap (Efron 1979; Efron and Tibshirani 1994)
- To use a parametric bootstrap approach, we would estimate the parameter on the observed data, simulate from the model with the estimated parameter, then re-estimate the parameter for each simulated realization
- The sampling distribution of the re-estimates is used as the depth-CD
- Theorem 1 in D. Liu, R. Y. Liu, and Min-ge Xie (2022) shows that this bootstrap distribution is an asymptotic depth-CD under certain regularity conditions
- For more info about the connections between bootstrap distributions and confidence distributions, see Example 3 in Min-ge Xie and Singh (2013)

Reason we use Gaussian depth-CD instead of bootstrap depth-CD

- We opt for an asymptotic Gaussian depth-CD, rather than a bootstrap depth confidence distribution, to improve the computational efficiency of our method
- This computational efficiency is particularly important, because our proposed method is intended to be applicable to models with intractable likelihoods
- In this setting, parameter estimation typically relies on simulation-based procedures that are substantially more computationally demanding than methods like maximum likelihood estimation
- Therefore, our procedure shares some of the motivations behind the Krinsky and Robb (1986) method

Estimating the asymptotic covariance

- Let $Q_n(\cdot)$ be sample discrepancy function s.t. $\hat{\theta} = \arg\min_{\theta \in \Theta} \|Q_n(\theta)\|$
- Let V be the asymptotic covariance matrix from $\sqrt{n} Q_n(\theta_0) \Rightarrow N(0, V)$
- Let $Q(\cdot)$ be population discrepancy function s.t. $Q_n(\theta) \xrightarrow{p} Q(\theta)$ uniformly over Θ
- Let Γ be the Jacobian matrix of $Q(\cdot)$ at θ_0
- We prove consistency $\hat{\theta} \xrightarrow{p} \theta_0$ and asymptotic normality

$$\sqrt{n}\left(\hat{\theta} - \theta_0\right) \Rightarrow N\left(0, (\Gamma^\top \Gamma)^{-1} \Gamma^\top V \Gamma (\Gamma^\top \Gamma)^{-1}\right)$$

- We need to estimate $(\Gamma^\top \Gamma)^{-1} \Gamma^\top V \Gamma (\Gamma^\top \Gamma)^{-1}$
- First, we estimate Γ using numerical differentiation tools (to be determined)
- Second, we estimate V by (slightly) extending covariance estimation results from nonstationary time series theory Mies and Steland (2023) and Mies (2023)
- For both estimators, use plug-in approach with $\hat{\theta}$ in place of the unknown θ_0

How to get synthetic training data for forecasting with TSFMs

- For forecasting with TSFMs, Dooley et al. (2023) pre-train by simulating from *multiplicative seasonality-trend-noise decomposition models* with many $\theta \in \Theta$
- We consider an extension with multiplicative autoregressive noise

$$Y_t = \mathrm{Se}\left(t, \theta^{Se}\right) \mathrm{Tr}\left(t, \theta^{Tr}\right) \exp\left(\kappa X_t - \frac{\kappa^2 \sigma_X^2}{2(1-\phi^2)}\right) \exp\left(\varepsilon_t^Y - \frac{\sigma_Y^2}{2}\right)$$
$$X_t = \phi X_{t-1} + \varepsilon_t^X$$

  for some scaling factor $\kappa > 0$ and autoregressive coefficient $\phi \in (-1, 1)$, with $X_0 \sim N(0, \frac{\sigma_X^2}{1-\phi^2})$ and $\varepsilon_t^X \overset{\text{iid}}{\sim} N(0, \sigma_X^2)$, $\varepsilon_t^Y \overset{\text{iid}}{\sim} N(0, \sigma_Y^2)$ for some $\sigma_X, \sigma_Y > 0$
- **Seasonality:** linear combination of sines, cosines (weekly, monthly, yearly periods)
- **Trend:** product of linear trend and exponential trend
- Our rolling-window estimator can be used (consistent and asymptotically normal)

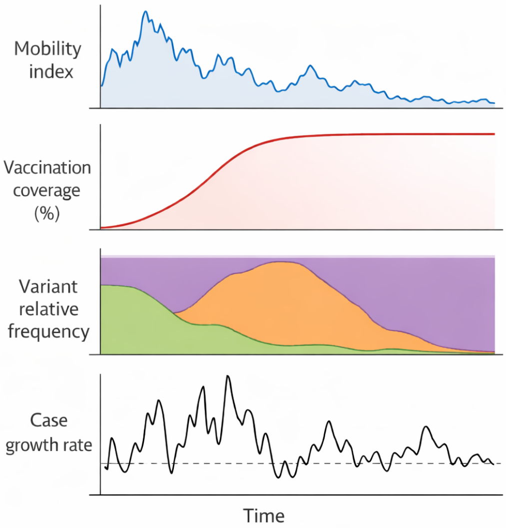
Model we use for smoothing application in epidemiology

- Discrete-valued process driven by a latent autoregressive process and controls
- We condition on the controls, so we can view each control as a discretely observed function of time $Z_t \equiv Z(t/n)$ that is sufficiently nice (e.g., α-Hölder, $\alpha > 1/2$)
- We define a (conditional) state-space model for $(X_{1:n}, Y_{1:n})$ given $Z_{1:n}$ as follows

$$
\begin{aligned}
Y_t \mid X_t, Z_t &\sim \mathrm{Pois}\left(\exp\left(\kappa X_t + \beta^\top Z_t\right)\right) \\
X_t &= \mu + \phi(X_{t-1} - \mu) + \varepsilon_t^X
\end{aligned}
$$

- Effect sizes $\kappa \in \mathbb{R}$, $\beta \in \mathbb{R}^{d_Z}$, autoregressive coefficient $\phi \in (-1, 1)$, and mean $\mu \in \mathbb{R}$, with $X_0 \sim N(\mu, \frac{\sigma_X^2}{1-\phi^2})$, $\varepsilon_t^X \overset{\text{iid}}{\sim} N(0, \sigma_X^2)$ indep. of $Z_{1:n}$, for some $\sigma_X > 0$
- $Y_{1:n}$ are observed case counts, $X_{1:n}$ is a *latent infectivity* process, and $Z_{1:n}$ are covariates (e.g., vaccination coverage, variant relative frequency, mobility)
- Our rolling-window estimator can be used (consistent and asymptotically normal)

COVID-19 time series May 2020–2022 (n=365x2=730, d=4)



Transformer architecture

- Our architecture is based on the MOIRAI architecture (Woo et al. 2024)
  – Note: Masked EncOder-based UnIveRsAl TIme Series Forecasting Transformer
- **Key idea:** uses “any-variate” attention which flattens the dimensions of a multivariate time series into a single sequence
- Theoretical investigations of MOIRAI in time series setting by D. Wu et al. (2025)
- We use the version of MOIRAI from D. Wu et al. (2025)

Setup

- Let $Y_t \in \mathbb{R}^{d_Y}$, $t \in \mathcal{T}_Y = \{1, \ldots, n\}$, and $Z_t \in \mathbb{R}^{d_Z}$, $t \in \mathcal{T}_Z \subseteq \{1, \ldots, n+H\}$, where $\mathcal{T}_Y \subseteq \mathcal{T}_Z$ because in the forecasting setting, the control variables may be known into the future for some maximal horizon $H \in \mathbb{N}$
- Denote $T_Y = |\mathcal{T}_Y|$ and $T_Z = |\mathcal{T}_Z|$
- Write $Y = (Y_t)_{t \in \mathcal{T}_Y} \in \mathbb{R}^{d_Y \times T_Y}$ and $Z = (Z_t)_{t \in \mathcal{T}_Z} \in \mathbb{R}^{d_Z \times T_Z}$
- The input to our transformer is the pair $(Y, Z) \in \mathbb{R}^{d_Y \times T_Y} \times \mathbb{R}^{d_Z \times T_Z}$
- Let $\mathcal{T}_{\text{output}} \subset \{1, \ldots, n+H\}$ and denote $T_{\text{output}} = |\mathcal{T}_{\text{output}}|$
- The output of our transformer is $\hat{\mathbf{q}} = (\hat{\mathbf{q}}_t)_{t \in \mathcal{T}_{\text{output}}} \in (\mathbb{R}^{d_X \times K})^{T_{\text{output}}}$
- In the smoothing setting, $\mathcal{T}_Z = \{1, \ldots, n\}$ and $T_{\text{output}} = \{1, \ldots, n\}$
- In the forecasting setting, $\mathcal{T}_Z = \{1, \ldots, n+H\}$ and $T_{\text{output}} = \{n+1, \ldots, n+H\}$

Any-variate encoding

- Define the dimensions by $D_p = d' + T_Z + 2$ and $D_e = d_Y + d_Z$
- As in Section 3.1 of D. Wu et al. (2025), for some $d' > 0$ for padding, let

$$p_t^Y = \begin{bmatrix} 0_{d'} \\ e_t^Y \\ 1 \\ \mathbb{I}(t \leq n) \end{bmatrix} \in \mathbb{R}^{D_p}, \quad p_t^Z = \begin{bmatrix} 0_{d'} \\ e_t^Z \\ 1 \\ \mathbb{I}(t \leq n) \end{bmatrix} \in \mathbb{R}^{D_p}$$

  where $e_t^Y, e_t^Z \in \mathbb{R}^{T_Z}$ are one-hot vectors with t-th time entry being 1 else 0
- As in Section 3.2 of D. Wu et al. (2025), $e_j^Y \in \mathbb{R}^{D_e}$, $e_\ell^Z \in \mathbb{R}^{D_e}$ are one-hot vectors with corresponding variable entry being 1 else 0 for $j \in [d_Y]$, $\ell \in [d_Z]$

Encoded input matrix

- For $T_Y = |\mathcal{T}_Y|$, $\mathcal{T}_Y = \{t_1, \ldots, t_{T_Y}\}$, the observed-series block is

$$H_Y = \begin{bmatrix} Y_{t_1,1} & \cdots & Y_{t_{T_Y},1} & \cdots & Y_{t_1,d_Y} & \cdots & Y_{t_{T_Y},d_Y} \\ p^Y_{t_1} & \cdots & p^Y_{t_{T_Y}} & \cdots & p^Y_{t_1} & \cdots & p^Y_{t_{T_Y}} \\ e^Y_1 & \cdots & e^Y_1 & \cdots & e^Y_{d_Y} & \cdots & e^Y_{d_Y} \end{bmatrix}$$

- For $T_Z = |\mathcal{T}_Z|$, $\mathcal{T}_Z = \{t_1, \ldots, t_{T_Z}\}$, the control-series block is

$$H_Z = \begin{bmatrix} Z_{t_1,1} & \cdots & Z_{t_{T_Z},1} & \cdots & Z_{t_1,d_Z} & \cdots & Z_{t_{T_Z},d_Z} \\ p^Z_{t_1} & \cdots & p^Z_{t_{T_Z}} & \cdots & p^Z_{t_1} & \cdots & p^Z_{t_{T_Z}} \\ e^Z_1 & \cdots & e^Z_1 & \cdots & e^Z_{d_Z} & \cdots & e^Z_{d_Z} \end{bmatrix}$$

Hidden-state notation

- Define

$$D = 1 + D_p + D_e, \quad N = d_Y T_Y + d_Z T_Z$$

- The any-variate encoding forms a matrix

$$H^{(0)} = [H_Y, H_Z] \in \mathbb{R}^{D \times N}$$

with one column for each scalar time/dimensional coordinate from Y and Z

- For each layer $\ell = 0, \ldots, \mathbf{L}$, let

$$H^{(\ell)} = [h_1^{(\ell)}, \ldots, h_N^{(\ell)}] \in \mathbb{R}^{D \times N}, \qquad h_r^{(\ell)} \in \mathbb{R}^D$$

Warm-up: Attention layer with ReLU (instead of softmax)

- A self-attention layer with M heads is denoted by $\mathrm{Attn}^{\dagger}_{\vartheta_0}(\cdot)$ with parameters $\vartheta_0 = \{(V_m), (Q_m), (K_m)\}_{m\in[M]}$ where $Q_m, K_m, V_m \in \mathbb{R}^{D\times D}$
- For any $H \in \mathbb{R}^{D\times N}$, define

$$\mathrm{Attn}^{\dagger}_{\vartheta_0}(H) = H + \frac{1}{N}\sum_{m=1}^{M}(V_m H)\,\sigma\Big((Q_m H)^{\top}(K_m H)\Big)$$

where

$$\sigma(t) = \frac{\mathrm{ReLU}(t)}{N}$$

Any-variate attention with ReLU (instead of softmax)

- An any-variate attention layer with M heads is denoted by $\mathrm{Attn}_{\vartheta_1}(\cdot)$ with parameters $\vartheta_1 = \{(V_m),(Q_m),(K_m),(u_m^1),(u_m^2)\}_{m\in[M]}$
- Let $U \in \mathbb{R}^{N\times N}$ be defined by:

$$U_{ij} = 1 \quad \Longleftrightarrow \quad \text{the } i\text{-th and } j\text{-th columns of } H \text{ correspond to the same variable}$$

and let $\bar{U} = I - U$

- Define any-variate attention as

$$\mathrm{Attn}_{\vartheta_1}(H) = H + \frac{1}{N}\sum_{m=1}^{M}(V_m H)\,\sigma\Big((Q_m H)^\top (K_m H) + u_m^1 * U + u_m^2 * \bar{U}\Big)$$

where $*$ denotes constant multiplication to all entries of a matrix

MLP layer

- An MLP layer with hidden dimension D' is denoted by

$$\mathrm{MLP}_{\vartheta_2}(\cdot)$$

with parameters

$$\vartheta_2 = (W_1, W_2), \qquad W_1 \in \mathbb{R}^{D' \times D}, \qquad W_2 \in \mathbb{R}^{D \times D'}$$

- For any

$$H \in \mathbb{R}^{D \times N}$$

define

$$\mathrm{MLP}_{\vartheta_2}(H) = H + W_2 \sigma(W_1 H)$$

Readout map

- All together, the **L** layers of the MOIRAI transformer are given by

$$\mathrm{TF}_{\vartheta}(H) = \mathrm{MLP}_{\vartheta_2^{\mathbf{L}}}\Big(\mathrm{Attn}_{\vartheta_1^{\mathbf{L}}}(\cdots \mathrm{MLP}_{\vartheta_2^1}(\mathrm{Attn}_{\vartheta_1^1}(H))\cdots)\Big)$$

- Let $\mathrm{read} : \mathbb{R}^{D\times N} \to \mathbb{R}^{d_X \times KT_{\mathrm{output}}}$ denote the readout map defined by

$$\mathrm{read}(H) = W_1^{\mathrm{read}} H W_2^{\mathrm{read}}$$

where $W_1^{\mathrm{read}} \in \mathbb{R}^{d_X\times D}$ and $W_2^{\mathrm{read}} \in \mathbb{R}^{N\times KT_{\mathrm{output}}}$

- Let $\mathrm{read}_t : \mathbb{R}^{D\times N} \to \mathbb{R}^{d_X\times K}$, $t \in \mathcal{T}_{\mathrm{output}}$, denote the subset of columns from the readout map which correspond to the prediction at time t, and denote

$$\hat{\mathbf{q}}_t = \mathrm{read}_t\big(\mathrm{TF}_{\vartheta}(H^{(0)})\big) \in \mathbb{R}^{d_X\times K}$$

- Denoting $\hat{\mathbf{q}} = (\hat{\mathbf{q}}_t)_{t\in\mathcal{T}_{\mathrm{output}}}$, our transformer is a map $(Y, Z) \mapsto \hat{\mathbf{q}}$

Parameters of MOIRAI transformer

- For $\ell \in [\mathbf{L}]$, the parameters corresponding to the ℓ-th layer are

$$\vartheta^\ell = [\vartheta_1^\ell, \vartheta_2^\ell], \quad \vartheta_1^\ell = [Q_m^\ell, K_m^\ell, V_m^\ell, u_m^{1,\ell}, u_m^{2,\ell}]_{m\in[M]}, \quad \vartheta_2^\ell = [W_1^\ell, W_2^\ell]$$

- The parameters of the readout map are

$$\vartheta^{\mathrm{read}} = [W_1^{\mathrm{read}}, W_2^{\mathrm{read}}]$$

- The full parameters of the MOIRAI transformer are given by

$$\vartheta = [\vartheta^1, \ldots, \vartheta^{\mathbf{L}}, \vartheta^{\mathrm{read}}]$$

Distributions of the stochastic system

- For all $\theta \in \Theta$ and $n \in \mathbb{N}$, denote by $P_{\theta,n}$ the distribution of $(G_t^{(n)}(\varepsilon_\ell, \theta))_{t\in[n],\ell\in\mathbb{Z}}$
- Distribution on $(\mathbb{R}^d)^{[n]\times\mathbb{Z}}$, i.e. the space of $\mathbb{R}^d$-valued functions of $[n] \times \mathbb{Z}$
- We observe the nonstationary time series $(G_t^{(n)}(\varepsilon_t, \theta))_{t\in[n]}$ at $\theta = \theta_0$
- At each fixed $t \in [n]$ and $\theta \in \Theta$, the process $(G_t^{(n)}(\varepsilon_\ell, \theta))_{\ell\in\mathbb{Z}}$ is stationary
- See the time-varying autoregressive process example

Example of nonstationary time series

- Time-varying autoregressive (tvAR(1)) process
- Given parameter curves $\beta : [0,1] \to (-1,1)$ and $\sigma : [0,1] \to \mathbb{R}$ define

$$X_t^{(n)} = \beta\left(\frac{t}{n}\right) X_{t-1}^{(n)} + \sigma\left(\frac{t}{n}\right) \varepsilon_t, \quad t = 1, \ldots, n$$

  where $(\varepsilon_t)_t$ are iid random variables
- Extend β, σ to domain $(-\infty, 1]$ by setting $\beta(u) = \beta(0)$ and $\sigma(u) = \sigma(0)$ for $u < 0$
- Substituting yields the time-varying moving average (tvMA(∞)) representation

$$X_t^{(n)} = \sum_{j=0}^{\infty} \phi_{t,j}^{(n)} \varepsilon_{t-j},$$

  where $\phi_{t,0}^{(n)} = \sigma\left(\frac{t}{n}\right)$ and for $j \geq 1$

$$\phi_{t,j}^{(n)} = \left(\prod_{r=0}^{j-1} \beta\left(\frac{t-r}{n}\right)\right) \sigma\left(\frac{t-j}{n}\right)$$

Nonstationarity and infill asymptotics

- When there is strong nonstationarity (i.e. not asymptotically stationary), we introduce rescaled time $u \in [0,1]$ and use infill asymptotics so observations at t/n
- **Intuition:** more observations at each local structure of the process as n grows
- Recall: $(\varepsilon_i)_{i\in\mathbb{Z}}$ is an iid sequence of random variables, we denote $\boldsymbol{\varepsilon}_t = (\varepsilon_t, \varepsilon_{t-1}, \ldots)$, and Θ is a parameter space
- For each $n \in \mathbb{N}$, let $G_t^{(n)} : \mathbb{R}^\infty \times \Theta \to \mathbb{R}^d$, $t = 1, \ldots, n$, and $\tilde{G}_u^{(n)} : \mathbb{R}^\infty \times \Theta \to \mathbb{R}^d$, $u \in [0,1]$, define a *generative model* that produces time series data

Assumption

Given a parameter $\theta \in \Theta$, index $n \in \mathbb{N}$ (linked to sample size and approximation), and noise inputs $\boldsymbol{\varepsilon}_t$, a time series $X_{1:n}$ is generated by nature as

$$X_t = G_t^{(n)}(\boldsymbol{\varepsilon}_t, \theta) = \tilde{G}_{t/n}^{(n)}(\boldsymbol{\varepsilon}_t, \theta), \quad t = 1, \ldots, n$$

Control of nonstationarity

- We measure the nonstationarity using a p-variation-type quantity, define

$$\left\| \left(\tilde{G}_u^{(i)}(\varepsilon_0, \theta) \right)_u \right\|_{p\text{-var}, \mathcal{L}^q(\theta)} = \sup_{\substack{0 = u_0 < \ldots < u_\ell = 1 \\ \ell \in \mathbb{N}}} \left(\sum_{j=1}^{\ell} \left\| \tilde{G}_{u_j}^{(i)}(\varepsilon_0, \theta) - \tilde{G}_{u_{j-1}}^{(i)}(\varepsilon_0, \theta) \right\|_{\mathcal{L}^q(\theta)}^p \right)^{\frac{1}{p}}$$

- We impose the following regularity condition to control the nonstationarity

Assumption

For some $q > 2$, some $p \in [1, 2)$, and all $i \in \mathbb{N}$, there exists a constant $\Lambda > 0$ such that

$$\sup_{\theta \in \Theta} \sup_{u \in [0,1]} \left\| \tilde{G}_u^{(i)}(\varepsilon_0, \theta) \right\|_{\mathcal{L}^q(\theta)} + \sup_{\theta \in \Theta} \left\| \left(\tilde{G}_u^{(i)}(\varepsilon_0, \theta) \right)_u \right\|_{p\text{-var}, \mathcal{L}^q(\theta)} \leq \Lambda$$

Replication and Rosenblatt transform

- We can take the random variables $(\varepsilon_\ell)_{\ell\in\mathbb{Z}}$ to be $U[0,1]$
- Standard results in probability theory imply that the causal representations based on measurable functions of sequences of iid $U[0,1]$ are already sufficiently general
- First, by Kallenberg (2021) Lemma 4.21, Lemma 4.22, and the surrounding discussion, we can replicate each of the iid $U[0,1]$ random variables by taking each $G_t^{(n)}$ to include compositions with measurable functions to replicate the original $U[0,1]$, which yields d iid $U[0,1]$ random variables $U_1,\ldots,U_d$
- Second, we can use the Rosenblatt transform (i.e. for each $j\in[d]$, sample $X_{t,j}$ from the conditional quantile given $X_{t,1},\ldots,X_{t,j-1}$ by transforming U_j), see Section 4 of W. B. Wu and Mielniczuk (2010)

Regression functions, errors, and residuals

- For each time t, we can always decompose the nonstationary time series as

$$X_t = f_t(Z_t) + \varepsilon_t, \quad Y_t = g_t(Z_t) + \xi_t$$

  where $f_t(z) = \mathbb{E}(X_t \mid Z_t = z)$, $g_t(z) = \mathbb{E}(Y_t \mid Z_t = z)$ are regression functions at time t, and the errors $(\varepsilon_t)_t$, $(\xi)_t$ can be dependent and nonstationary
- Denote the vectorized outer product of the errors at time t by $R_t = \mathrm{vec}(\varepsilon_t \xi_t^\top)$
- Let $\hat{f}_t$ and $\hat{g}_t$ be estimates of f_t and g_t, respectively, and denote the residuals by

$$\hat{\varepsilon}_t = X_t - \hat{f}_t(Z_t), \quad \hat{\xi}_t = Y_t - \hat{g}_t(Z_t)$$

- Denote the vectorized outer product of the residuals at time t by $\hat{R}_t = \mathrm{vec}(\hat{\varepsilon}_t \hat{\xi}_t^\top)$

Regression estimators for nonstationary time series

- Recall: $\mathcal{P}_n^* \subset \mathcal{P}_n^{\mathrm{CI}}$ is collection of null distributions $\mathcal{P}_n^{\mathrm{CI}}$ s.t. *all assumptions hold*
- For all distributions $P \in \mathcal{P}_n^*$, we require the estimators of the regression functions to satisfy the following conditions:
  1. The **product** of $L^2(P)$ norms of the estimation errors must be $o\left(1/\sqrt{n}\,\mathrm{polylog}(n)\right)$
  2. Each $L^2(P)$ norm only needs to be $o\left(1/\mathrm{polylog}(n)\right)$
- The exact notation is given on the next slide
- Many regression estimators for nonstationary time series have been developed
- **Examples:** linear regression, kernel smoothing, sieve, boosting, neural networks
- Rescaled time $t/n \in [0,1]$ is included as an argument along with the covariates

Required convergence rates for regression estimators

- Denote the estimation errors at time t by

$$\Delta^f_{P,t,i,a} = f_{P,t,i,a}(Z_t) - \hat{f}_{t,i,a}(Z_t), \quad \Delta^g_{P,t,j,b} = g_{P,t,j,b}(Z_t) - \hat{g}_{t,j,b}(Z_t)$$

- For each combination (i, j, a, b) of dimensions $i \in [d_X]$, $j \in [d_Y]$ and time-offsets $a \in A_i$, $b \in B_j$, under consideration, we have

$$\sup_{P \in \mathcal{P}_n^*} \sup_{i \in [d_X], a \in A_i} \sup_{t \in [n]} \mathbb{E}_P \left(\left| \Delta^f_{P,t,i,a} \right|^2 \right)^{1/2} = o\left(1/\mathrm{polylog}(n)\right)$$

$$\sup_{P \in \mathcal{P}_n^*} \sup_{j \in [d_Y], b \in B_j} \sup_{t \in [n]} \mathbb{E}_P \left(\left| \Delta^g_{P,t,j,b} \right|^2 \right)^{1/2} = o\left(1/\mathrm{polylog}(n)\right)$$

$$\sup_{P \in \mathcal{P}_n^*} \sup_{(i,j,a,b)} \sup_{t \in [n]} \mathbb{E}_P \left(\left| \Delta^f_{P,t,i,a} \right|^2 \right)^{1/2} \mathbb{E}_P \left(\left| \Delta^g_{P,t,j,b} \right|^2 \right)^{1/2} = o\left(1/\sqrt{n}\ \mathrm{polylog}(n)\right)$$

Main ideas of distribution-uniform strong Gaussian approximation

- Based on strong Gaussian approximation from Mies and Steland (2023)
- Suppose the aforementioned assumptions hold for some collection of distributions $\mathcal{P}_n^*$ for the time series $(W_t)_{t\in[n]}$
    1. Representation in terms of noise inputs
    2. Decay of physical dependence measure
    3. Control of nonstationarity
- **Main idea:** Uniformly over a collection of distributions, we can closely couple $\sqrt{n}$-scaled partial sums of $(W_t)_t$ and a nonstationary Gaussian process
- A more precise statement is given on the next slide, for when the constant $\rho \geq 3$ for the decay of the physical dependence measure, and when $q > 2$ is large

More details for distribution-uniform strong Gaussian approximation

- On a potentially enriched collection of probability spaces $(\Omega, \mathcal{F}, \mathbb{P}_P)_{P \in \mathcal{P}_n^*}$ indexed by the pushforward measures $P \in \mathcal{P}_n^*$, there exist random vectors $(W_t')_{t \in [n]}$ with the same distribution as $(W_t)_{t \in [n]}$ for each $P \in \mathcal{P}_n^*$, and *independent* Gaussian random vectors $(V_t')_{t \in [n]}$ where $V_t' \sim N(0, \Sigma_{P,t})$ for all $P \in \mathcal{P}_n^*$ and $t \in [n]$, s.t.

$$\sup_{P \in \mathcal{P}_n^*} \left(\mathbb{E}_P \max_{T \in [n]} \left\| \frac{1}{\sqrt{n}} \sum_{t \leq T} (W_t - V_t') \right\|^2 \right)^{\frac{1}{2}} = o\left(\frac{\mathrm{polylog}(n)}{n^{\frac{1}{6}}} \right)$$

where $\Sigma_{P,t}$ is the *local long-run* covariance matrix at time t defined by

$$\Sigma_{P,t} = \Sigma_{P_\theta,t}^{(n)} = \sum_{h=-\infty}^{\infty} \mathrm{Cov}_{P_\theta} \left(G_t^{(n)}(\varepsilon_0, \theta), G_t^{(n)}(\varepsilon_h, \theta) \right)$$

Choices of features

- **Method of simulated moments:** F based on functions of moments
  - Sample mean, variance, skewness, autocorrelation, and so on
- **Indirect inference:** F from parameter estimates in auxiliary model
  - To get $\hat{\theta}$ for MA(1): $X_t = \varepsilon_t + \theta_0 \varepsilon_{t-1}$
  - Use $F = \hat{\beta}$ from AR(1): $X_t = \beta X_{t-1} + \xi_t$
- **Synthetic likelihood:** F from user-chosen summary statistics
  - Coefficients from polynomial regression of X_t on $X_{t-1}, \ldots, X_{t-\ell}$
- **Neural summary statistics:** F from neural network
  - Extract early layer of neural network

Random Fourier features

- Let $x_1, \ldots, x_{m+1}$ be vectors in $\mathbb{R}^d$, and define $x = (x_1, \ldots, x_{m+1})^\top \in \mathbb{R}^{(m+1)\times d}$
- We focus on random Fourier features, though our framework can be used with different random features
- Specifically, we consider random Fourier features of the form

$$\varphi_i(x) = \cos\left(\sum_{j=1}^{m+1} \Omega_{i,j} \cdot x_j + \alpha_i\right)$$

where $\Omega_{i,j} \overset{\text{iid}}{\sim} N(0, I_d)$ and $\alpha_i \overset{\text{iid}}{\sim} U(-\pi, \pi)$ for $i = 1, \ldots, k$ and $j = 1, \ldots, m+1$

- Denote all k of these functions by

$$\varphi = (\varphi_1, \ldots, \varphi_k)$$

so that $\varphi : \mathbb{R}^{(m+1)\times d} \to \mathbb{R}^k$

Random Fourier features

- Define the i-th random feature of the observed and r-th simulated time series at time $t = m+1, \ldots, n$ as

$$f_{t,i}^{\text{obs}} = \varphi_i(X_{t-m:t}^{\text{obs}}), \quad f_{t,i}^{(r)}(\theta) = \varphi_i\left(X_{t-m:t}^{(r)}(\theta)\right)$$

- Write all k random features at time t as

$$f_t^{\text{obs}} = \left(f_{t,1}^{\text{obs}}, \ldots, f_{t,k}^{\text{obs}}\right), \quad f_t^{(r)}(\theta) = \left(f_{t,1}^{(r)}(\theta), \ldots, f_{t,k}^{(r)}(\theta)\right)$$

Time-average estimator

- Define the observed and simulated time-averages of the random features by

$$F^{\mathrm{obs}} = \frac{1}{n-m}\sum_{t=m+1}^{n} f_t^{\mathrm{obs}}, \quad \bar{F}^{\mathrm{sim}}(\theta) = \frac{1}{n-m}\sum_{t=m+1}^{n} \bar{f}_t^{\mathrm{sim}}(\theta)$$

where $\bar{f}_t^{\mathrm{sim}}(\theta) = \frac{1}{s}\sum_{r=1}^{s} f_t^{(r)}(\theta)$. The time-average estimator is given by

$$\hat{\theta}^{\mathrm{TA}} = \operatorname*{argmin}_{\theta\in\Theta} \left\| \hat{Q}_n^{\mathrm{TA}}(\theta) \right\|$$

where the sample discrepancy function is defined as

$$\hat{Q}_n^{\mathrm{TA}}(\theta) = F^{\mathrm{obs}} - \bar{F}^{\mathrm{sim}}(\theta)$$

- In practice, an optimization procedure is used to find the minimizer of $\left\| \hat{Q}_n^{\mathrm{TA}}(\cdot) \right\|$

Rolling-window estimator

- Define the observed and simulated rolling-window random features at time t by

$$F_t^{\mathrm{obs}} = \frac{1}{w \wedge t} \sum_{j=(t-w)\vee 1}^{t} f_j^{\mathrm{obs}}, \quad \bar{F}_t^{\mathrm{sim}}(\theta) = \frac{1}{w \wedge t} \sum_{j=(t-w)\vee 1}^{t} \bar{f}_j^{\mathrm{sim}}(\theta),$$

where $\bar{f}_t^{\mathrm{sim}}(\theta) = \frac{1}{s}\sum_{r=1}^{s} f_t^{(r)}(\theta)$ and $w \in \mathbb{N}$ is the window size

Rolling-window estimator

- The rolling-window estimator is given by

$$\hat{\theta}^{\mathrm{RW}} = \underset{\theta \in \Theta}{\operatorname{argmin}} \left| \hat{Q}_n^{\mathrm{RW}}(\theta) \right|,$$

where the sample discrepancy function is defined as

$$\hat{Q}_n^{\mathrm{RW}}(\theta) = \frac{1}{n-m} \sum_{t=m+\tau+L}^{n} \left[\left\| F_t^{\mathrm{obs}} - \bar{F}_t^{\mathrm{sim}}(\theta) \right\|^2 + K_t(\theta) \right],$$

$$K_t(\theta) = 2 \left(F_{t-L}^{\mathrm{obs}} - \bar{F}_{t-L}^{\mathrm{sim}}(\theta) \right)^\top \left(\left[f_t^{\mathrm{obs}} - \bar{f}_t^{\mathrm{sim}}(\theta) \right] - \left[F_{t-L}^{\mathrm{obs}} - \bar{F}_{t-L}^{\mathrm{sim}}(\theta) \right] \right),$$

where $\tau \in \mathbb{N}$ is an initial time-offset and $L \in \mathbb{N}$ is a lag

- In practice, an optimization procedure is used to find the minimizer of $\left| \hat{Q}_n^{\mathrm{RW}}(\cdot) \right|$

What do we mean by "almost every" function? Prevalent subset

Definition (J. Yorke and Ott (2005))

Let V be a completely metrizable topological vector space. A Borel set $E \subset V$ is said to be **prevalent** if there exists some Borel measure μ on V such that:

1. $0 < \mu(C) < \infty$ for some compact subset C of V
2. The set $E + v$ has full μ-measure (that is, the complement of $E + v$ has measure zero) for all $v \in V$

- In this literature, it is common to say that, if $E \subset V$ contains a prevalent Borel set, then "almost every" element of V lies in E
- In this sense, "almost every" C^1 smooth map from $\mathbb{R}^p$ to $\mathbb{R}^{2p+1}$ has the properties from the embedding theorem of Sauer, J. A. Yorke, and Casdagli (1991)

Definition of smooth manifold (University of Toronto Lecture Notes)

Definition

A C^∞-manifold (i.e. smooth manifold or differentiable manifold) is a Hausdorff topological space M, with countable basis, together with a C^∞-atlas

- The next slides explain the definitions needed to understand this

Definition of smooth manifold (University of Toronto Lecture Notes)

- An n-dimensional manifold is a space that locally looks like $\mathbb{R}^n$. To give a precise meaning to this idea, our space first of all has to come equipped with some topology (so that the word "local" makes sense)
- Recall that a *topological space* is a set M, together with a collection of subsets of M, called *open subsets*, satisfying the following three axioms: (i) the empty set $\emptyset$ and the space M itself are both open, (ii) the intersection of any finite collection of open subsets is open, (iii) the union of any collection of open subsets is open
- The collection of open subsets of M is also called the topology of M. A map $f : M_1 \to M_2$ between topological spaces is called *continuous* if the pre-image of any open subset in M_2 is open in M_1. A continuous map with a continuous inverse is called a *homeomorphism*

Definition of smooth manifold (University of Toronto Lecture Notes)

- One ingredient in the definition of a manifold is that our topological space comes equipped with a covering by open sets which are homeomorphic to open subsets of $\mathbb{R}^n$
- Let M be a topological space. An n-dimensional *chart* for M is a pair (U, ϕ) consisting of an open subset $U \subset M$ and a continuous map $\phi : U \to \mathbb{R}^n$ such that ϕ is a homeomorphism onto its image $\phi(U)$. Two such charts (U_α, ϕ_α), (U_β, ϕ_β) are C^∞-compatible if the transition map

$$\phi_\beta \circ \phi_\alpha^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \to \phi_\beta(U_\alpha \cap U_\beta)$$

is a diffeomorphism (a smooth map with smooth inverse). A covering $\mathcal{A} = (U_\alpha)_{\alpha \in A}$ of M by pairwise C^∞-compatible charts is called a C^∞-atlas

Definition of smooth manifold (University of Toronto Lecture Notes)

- We recall that a topological space is called *Hausdorff* if any two points have disjoint open neighborhoods.
- A *basis* for a topological space M is a collection $\mathcal{B}$ of open subsets of M such that every open subset of M is a union of open subsets in the collection $\mathcal{B}$.
- A topological space with countable basis is also called *second countable*
- For example, the collection of open balls $B_\varepsilon(x)$ in $\mathbb{R}^n$ define a basis. But one already has a basis if one takes only all balls $B_\varepsilon(x)$ with $x \in \mathbb{Q}^n$ and $\varepsilon \in \mathbb{Q}_{>0}$; this defines a countable basis.

References I

- Adler, Coen et al. (2025). "Beyond accuracy: Are time series foundation models well-calibrated?" arXiv preprint arXiv:2510.16060.
- Bodik, Juraj and Olivier C. Pasche (2024). "Granger causality in extremes". arXiv preprint arXiv: 2407.09632.
- Dooley, Samuel et al. (2023). "Forecastpfn: Synthetically-trained zero-shot forecasting". In: *Advances in Neural Information Processing Systems* 36, pp. 2403–2426.
- Efron, Bradley (1979). "Bootstrap methods: Another look at the jackknife". In: *Annals of Statistics* 7.1, pp. 1–26.
- Efron, Bradley and Robert J. Tibshirani (1994). *An introduction to the bootstrap*. Chapman and Hall/CRC.
- Kallenberg, Olav (2021). *Foundations of modern probability*. Third edition. Springer.

References II

- Krinsky, Itzhak and A. Leslie Robb (1986). "On approximating the statistical properties of elasticities". In: *The Review of Economics and Statistics*, pp. 715–719.
- Liu, Dungang, Regina Y. Liu, and Min-ge Xie (2022). "Nonparametric fusion learning for multiparameters: Synthesize inferences from diverse sources using data depth and confidence distribution". In: *Journal of the American Statistical Association* 117.540, pp. 2086–2104.
- Liu, Regina Y., Jesse M. Parelius, and Kesar Singh (1999). "Multivariate analysis by data depth: descriptive statistics, graphics and inference". In: *The Annals of Statistics* 27.3, pp. 783–858.
- Liu, Xu et al. (2025). "Empowering time series analysis with synthetic data: A survey and outlook in the era of foundation models". arXiv preprint arXiv:2503.11411.
- Mahalanobis, Prasanta Chandra (1936). "On the generalized distance in statistics". In: *Proceedings of the National Institute of Science of India* 2, pp. 49–55.

References III

- Mies, Fabian (2023). "Functional estimation and change detection for nonstationary time series". In: *Journal of the American Statistical Association* 118.542, pp. 1011–1022.
- Mies, Fabian and Ansgar Steland (2023). "Sequential Gaussian approximation for nonstationary time series in high dimensions". In: *Bernoulli* 29.4, pp. 3114–3140.
- Sauer, Tim, James A. Yorke, and Martin Casdagli (1991). "Embedology". In: *Journal of Statistical Physics* 65.3, pp. 579–616.
- Shah, Rajen D. and Jonas Peters (2020). "The hardness of conditional independence testing and the generalised covariance measure". In: *Annals of Statistics* 48.3, pp. 1514–1538.
- Woo, Gerald et al. (2024). "Unified training of universal time series forecasting transformers". In: *International Conference on Machine Learning* 41.
- Wu, Dennis et al. (2025). "Transformers and their roles as time series foundation models". arXiv preprint arXiv:2502.03383.

References IV

Wu, Wei Biao (2005). "Nonlinear system theory: another look at dependence". In: *Proceedings of the National Academy of Sciences* 102.40, pp. 14150–14154.

Wu, Wei Biao and Jan Mielniczuk (2010). "A new look at measuring dependence". In: *Dependence in Probability and Statistics*, pp. 123–142.

Xie, Min-ge and Kesar Singh (2013). "Confidence distribution, the frequentist distribution estimator of a parameter: A review". In: *International Statistical Review* 81.1, pp. 3–39.

Yorke, James and William Ott (2005). "Prevalence". In: *Bulletin of the American Mathematical Society* 42.3, pp. 263–290.